

Reading

General Strategy & Tips:

- Use QAPE
 - Question
 - Analyze
 - Predict
 - Eliminate
 - Ignore the fluff
 - Get excited about transitions and words of negation
 - Put things in your own words
-

Words in Context:

- Look for *synonyms or antonyms* for the blank space in the text
- Treat word replacement like a blank space (ignore the original word)
- Don't choose unknown words unless you have eliminated 3 wrong answers

Central Ideas and Details:

- Non-fiction main ideas tend to live at the *beginning and end* of passages
- Fiction main ideas emerge from *context and synthesis* across the passage
- For detail-based questions, just skim the passage looking for keywords (or synonyms of keywords)

Structure & Purpose:

- Purpose is more about what the text *does*, and answers are often *general (not specific)*
- Structure is simply breaking the text into its different parts in order
- Consider the function of an underlined phrase in relation to the sentence before and after it

Inferences:

- Try reading the last sentence first to prime your focus
- You'll likely end up reading the whole passage

Command of Evidence:

- Find the claim and put it in your own words
- Quantitative: Ignore the graphic unless you *must* look at it
- Textual: Ignore the rest of the passage unless you need to clarify something

Cross-Text Connections:

- If a text is responding or describing
 - Start by identifying the point of view (main idea) of the text that will be *responding*.
 - Now read the other text, pretending *you* have the opinion of the first text you read.
 - If you're looking for what the texts would agree on
 - Identify the main idea of each text, but also keep an eye out for any *details* that are consistent in both texts.
 - Look for specifics to save time
 - You might end up reading everything, but sometimes you find what you need quickly using specific words or names from the question stem.
-

Common Distractor Answers:

- Conclusion / Recommendation (for main idea)
- Too broad / Too narrow
- Out-of-scope
- Opposite
- Extreme or Absolute
- Distorted
- No Comparison (for command of evidence)



General Strategy & Tips:

- Use **Are**
 - Answers
 - Rules
 - Eliminate
- Don't read everything
- Read out loud (in your head)
- Trust grammar rules over "what sounds right"

Sentence Structure:

- Every complete sentence needs a subject + verb and cannot start with a dependent clause
- Comma splices (commas acting like periods) are everywhere—don't choose them!

Apostrophes (Possession & Contractions):

- Singular possession
 - Normal: Julianna's cat is really mean!
 - Ending in s: James's cat is really sweet!
- Plural possession
 - Ending in s: The cats' tails are so fluffy.
 - Irregular: The children's cats have no right to my dinner.
- Plural individuals
 - Individual: Maria's and Omar's cats are both beautiful.
 - Shared: Robert and Natalie's cat is ugly.
- Contractions: I'm, they've, dad's, can't, won't, don't, shouldn't, couldn't, etc.
 - It's = it + is. Its = possessive.
 - There = location/existence. Their = possessive. They're = they are.
 - Who's = who is. Whose = possessive.

Pronouns:

- Singular subjects → he/him, she/her, it.
- Plural subjects → they/them, we/us.
- Those/these (plural) and that/this (singular) are usually distractors.
- Choose one/you only if it already appears in the passage (or after eliminating the others).

Modifiers:

- Whatever/whoever is described before the comma must come right after the comma.
 - Wrong: Eager to impress his boss..., Dave's unfortunate *decision* was...
 - Right: Eager to impress his boss..., *Dave* made the unfortunate decision...
- Use the Front Flip trick

Verbs:

- Subject–Verb Agreement (most questions)
 - Singular subject → singular verb; plural subject → plural verb.
 - Use the odd verb out trick
- Tense Agreement (fewer questions)
 - Keep tense consistent unless context requires a shift.
- Infinitive Use (rare)
 - Know the "to + verb" form and choose it when the sentence demands it.

Rhetorical Synthesis

- Identify the goal in the question (e.g., emphasize a similarity, explain an advantage, present a primary aim).
- Eliminate choices that don't hit that goal directly:
 - Mentions one thing when the task requires a comparison → cut it.
 - Gives a conclusion when the task asks for a purpose → cut it.
- Check the notes only if needed (multiple answers seem right or you need clarification).
- The best answer hits the goal directly—nothing extra, nothing missing.



Writing

Punctuation:

Commas

1. Separate items in a list
 - My favorite foods are ice cream, pizza, and mangoes.
2. Set off non-essential info
 - Eating, which is my favorite activity, is something we all must do.
3. Divide independent clauses with FANBOYS
 - I told a chemistry joke, but there was no reaction.
4. Conclude an intro word, phrase, or clause
 - After correcting my pronunciation, Hermione nailed the spell with ease.

Colons

1. Must come after an independent clause
 - *Wrong*—My favorite foods are: ice cream, pizza, and mangoes.
 - *Correct*—My favorite foods are ice cream, pizza, and mangoes.
2. Introduce lists
 - I have three favorite foods: ice cream, pizza, and mangoes.
3. Give an example
 - My roommate is terrible at cooking: last night he burned water.
4. Resolve cliffhangers
 - Bob's biggest fear came true: he was out of coffee.

Semicolons

1. Act like periods (90% of the time)
 - I told a chemistry joke; there was no reaction.
 - I don't like pizza; I love it!
2. Separate items in a complex list
 - My favorite foods are ice cream, which is cold and comforting; pizza, which is hot and also comforting; and mangoes, which are downright magical.

Dashes

- Set off non-essential info
 - Eating—which is my favorite activity—is something we all must do.
- Steal the power from other punctuation marks
 - I told a chemistry joke—but there was no reaction. (comma)
 - I don't like pizza—I love it! (semicolon/period)
 - Bob's biggest fear came true—out of coffee again! (colon)
 - He walked to the bathroom, took out his phone—and threw it in the toilet. (comma/ellipsis)

Transitions:

- Use the SLAP strategy:
 - **Scan** – Read the sentence before and the one with the blank
 - **Link** – Identify the relationship (contrast, cause-effect, continuation, etc.)
 - **Anticipate** – Think of a word that fits (e.g., “however” for contrast)
 - **Purge** – Eliminate choices that don't match



Common Transitions

Transition Type	Transition Words	Sample Sentence	Interchangeability
Additions or Continuations (Seamlessly adding similar information)	Additionally, Furthermore, In addition, Moreover, Also	The company implemented new training programs. Additionally, it updated its safety protocols.	Interchangeable in most cases
Comparisons (Highlighting similarity or parallel ideas)	Likewise, Similarly	The museum expanded its exhibits to attract new visitors. Likewise, the library increased its community outreach.	Interchangeable in most cases
Examples (Providing specific examples or instances)	For example, For instance	Many species are endangered due to habitat loss. For example, the Amur leopard has seen a sharp decline in population.	Interchangeable
Specific Emphasis (Highlighting particular aspects of a topic)	Specifically, In particular	Pollution affects many habitats. Specifically, the coral reefs are at risk due to rising ocean temperatures.	Unique-Use (Different emphases)
Clarifications or Rephrasings (Restating ideas for clarity)	In other words, That is, That is to say	Her popularity was nonexistent. In other words, nobody liked her.	Interchangeable in some cases
Summative Conclusions (Concluding or summarizing ideas)	In sum, In summary	(Insert long explanation here). In summary, it takes a long time to plan a webinar.	Interchangeable in all cases
Contrasts or Concessions (Presenting a counterpoint or acknowledging an exception)	However, Nevertheless, Nonetheless, Still, Granted, That being said, Admittedly, Regardless	He wanted to invest in tech stocks. However, his advisor recommended more conservative options.	Interchangeable in many cases



Common Transitions

Transition Type	Transition Words	Sample Sentence	Interchangeability
Alternatives or Options (Presenting alternative perspectives or choices)	On the other hand, In contrast, By contrast, Conversely, Alternately, Alternatively, In comparison, Instead	The company could expand its workforce. On the other hand, it could focus on training existing employees.	Unique-Use (Indicates specific options or contrasts)
Sequence or Order (Indicating order or progression in steps)	First, Next, Then, Subsequently, Finally, Ultimately, Secondly	First, preheat the oven. Next, mix the ingredients.	Unique-Use (Each indicates a specific step)
Cause and Effect—Direct Result (Explaining direct outcomes)	As a result, Consequently, Therefore, Thus, Hence	The team missed the deadline. As a result, they had to work overtime.	Interchangeable in many cases
Cause and Effect—Purpose or Goal-Oriented (Linking to a goal, outcome, or purpose)	For this reason, Accordingly, To that end	The project required more resources. For this reason, the team requested additional funding.	Interchangeable in many cases
Emphasis or Intensification (Highlighting importance or significance)	In fact, Indeed	Her work has been groundbreaking. In fact, she recently won an award for innovation.	Interchangeable in many cases
Time References (Referring to timing or temporal shifts)	Earlier, Previously, Meanwhile, Today, Currently, Later, Nowadays	The company struggled to meet demands last year. Today, it has expanded its workforce to keep up.	Unique-Use (Each term signals a specific time)
Context-Based Transitions (Flexible transitions adaptable to various purposes based on sentence context)	Actually, Elsewhere, In many cases, In reality, In turn, Fittingly	The new policy may be challenging to implement. Actually, it requires extensive training.	Flexible Use (Adaptable based on context)

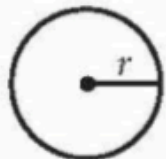


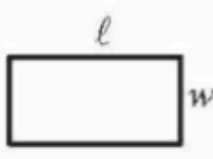
DESMOS SAT KEYBOARD SHORTCUTS

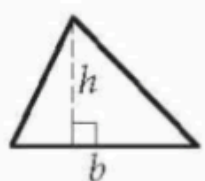
Function	Shortcut	Example
Exponent	<code>^</code> (Shift + 6)	$x^3 \rightarrow x^3$
Less Than or Equal To/ Greater Than or Equal To	<code><=</code> <code>>=</code>	$a \geq b \rightarrow a \geq b$ $a \leq b \rightarrow a \leq b$
Fraction / Division	<code>frac</code> or <code>/</code>	$x/6 \rightarrow \frac{x}{6}$
Square Root	<code>sqrt</code>	$\text{sqrt}64 \rightarrow \sqrt{64}$
Cube Root	<code>cbrt</code>	$\text{cbrt}27 \rightarrow \sqrt[3]{27}$
Nth Root	<code>nthroot</code>	$\text{nthroot}(5, 200) \rightarrow \sqrt[5]{200}$
Multiplication	<code>*</code>	$4 * 3 \rightarrow 4 \cdot 34$
Absolute Value	<code> </code> (Shift + \) or <code>abs()</code>	$\text{abs}(x-3) \rightarrow x-3 $
Pi	<code>pi</code>	$\text{pi} \rightarrow \pi$

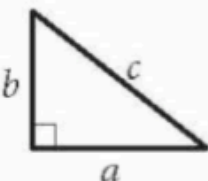
Bluebook “Reference” Sheet

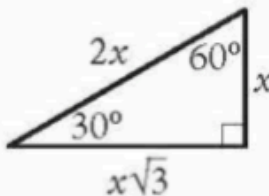
REFERENCE


$$A = \pi r^2$$
$$C = 2\pi r$$

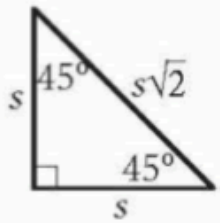

$$A = \ell w$$

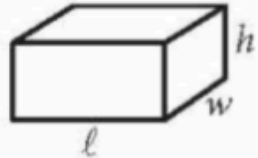

$$A = \frac{1}{2}bh$$

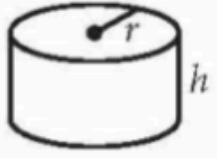

$$c^2 = a^2 + b^2$$

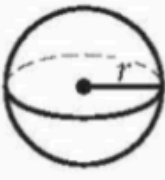


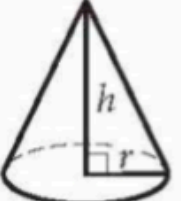
Special Right Triangles

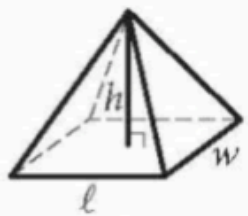



$$V = \ell wh$$


$$V = \pi r^2 h$$


$$V = \frac{4}{3}\pi r^3$$


$$V = \frac{1}{3}\pi r^2 h$$


$$V = \frac{1}{3}\ell wh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of the measures in degrees of the angles of a triangle is 180.

Remember, this shows the radian conversion rate!

2π radians = 360 degrees
so π radians = 180 degrees

FORMS OF LINEAR EQUATIONS

Standard Form: $Ax + By = C$

- *x-intercept: $(C/A, 0)$*
- *y-intercept: $(0, C/B)$*

Slope-Intercept Form: $y = mx + b$

Slope
y-intercept

Point-Slope Form: $y - y_1 = m(x - x_1)$

- *Useful if you know the slope and one point*
- *Converts into the slope equation when we isolate m : $m = \frac{y_1 - y_2}{x_1 - x_2}$*

- **Slope (Rate of Change):** $m = \frac{y_1 - y_2}{x_1 - x_2}$
- **Parallel Lines:** same slope
- **Perpendicular Lines:**
 - Negative reciprocal slopes (for example, $\frac{5}{2}$ and $-\frac{2}{5}$)
 - Product of slopes = -1
- **Systems of Equations:**
 - Solution: Where two lines meet
 - One solution: lines intersect once
 - No solution: parallel lines
 - Infinite solutions → same line
- **Word Problems on SAT:**
 - Slope-intercept form
 - y-intercept (b): starting point
 - slope (m): constant rate of change
 - y: total
 - x: affects the total (often time or # of items)
 - Standard Form
 - x & y: items
 - A & B: values
 - C: total



FORMS OF QUADRATICS

Standard Form: $y = ax^2 + bx + c$

c is the y-intercept

Factored Form: $y = a(x - p)(x - q)$

The solutions (x-intercepts) are p and q

Vertex Form: $y = a(x - h)^2 + k$

(h, k) is the vertex

- A “solution”
 - For a quadratic: where the parabola touches the x-axis
 - For a system: where the equations touch each other
- Axis of Symmetry (x-coordinate of the vertex)
 - Standard Form: $\frac{-b}{2a}$
 - Factored Form: $\frac{p+q}{2}$
- Sum of Solutions
 - Standard Form: $\frac{-b}{a}$
 - Factored Form: $p + q$
 - Vertex Form: $2h$
- Product of Solutions
 - Standard Form: $\frac{c}{a}$
 - Factored Form: pq
- Quadratic Formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- For all forms, “a” controls direction and width of the parabola
- “Minimum” and “Maximum” refer to the y-coordinate of the vertex



PERCENTAGES

1. PERCENT OF

To calculate the percent “of” something, **multiply by the decimal version of the percent.**

Example: What is 30% of 10?

30% -----> .3

$$10(.3) = 3$$

2. PERCENT CHANGE

To calculate the increase or decrease in percent (% change), **use this formula:**

$$\% \text{ change} = \frac{\text{Difference}}{\text{Original}}$$

← difference between original and new value
↖ original value

Example: Last year I had \$4000 in my savings. Now I have \$6500. By what % did my savings increase?

$$\frac{6500 - 4000}{4000} = \frac{2500}{4000} = .625 = 62.5\%$$

3. ADDING PERCENT

To add %, multiply by 1 plus the % (in decimal form of course).

So to add 35% to 100, we would do ...

Original 100%

$$100(1.35)$$

↖ additional 35%

4. SUBTRACTING PERCENT

To subtract %, multiply by 100% minus the % (in decimal form of course).

So to subtract 35% from 100, we would do ...

$$100(0.65)$$

↖ 100 - 35 is 65



TURNING WORDS INTO MATH

Equals (=)

- is, was, will be, equals, results in, gives
- Example: The total is 25 $\rightarrow x + 7 = 25$

Addition (+)

- sum, more than, increased by, combined, altogether
- Example: A number increased by 6 $\rightarrow x + 6$
- Example: The sum of a number and 12 is 20 $\rightarrow x + 12 = 20$

Subtraction (−)

- difference, less than, decreased by, minus, fewer than, take away
- Example: A number decreased by 8 $\rightarrow x - 8$
- Example: Seven less than a number $\rightarrow x - 7$

Multiplication (×)

- product, times, twice, triple, double, of (especially in percents)
- Example: Twice a number $\rightarrow 2x$
- Example: 30% of 50 $\rightarrow 0.3(50)$

Division (÷ or /)

- quotient, per, ratio of, out of, each, every, split among
- Example: The ratio of boys to girls is 3 to 4 $\rightarrow 3/4$
- Example: 50 students split among 5 teams $\rightarrow 50 \div 5 = 10$
- Example: Each student gets 2 pencils $\rightarrow \text{pencils} \div \text{students}$

Inequalities

- Greater than ($>$): more than, exceeds
 - $x > 10 \rightarrow$ The number exceeds 10
- Less than ($<$): fewer than, under, smaller than
 - $x < 5 \rightarrow$ The number is fewer than 5
- Greater than or equal ($\geq$): at least, no less than
 - $x \geq 18 \rightarrow$ You must be at least 18 years old
- Less than or equal ($\leq$): at most, no more than
 - $x \leq 100 \rightarrow$ The speed is no more than 100 mph

ADDITIONAL GEOMETRY FORMULAS

Circles

- Arc Length = $(\theta/360)(2\pi r)$
- Sector Area = $(\theta/360)(\pi r^2)$
- Equation of a circle: $(x - h)^2 + (y - k)^2 = r^2$

Triangles

- Exterior angle = sum of the two remote interior angles
- Triangle inequality: sum of any two sides $>$ third side
- Isosceles: two equal sides $\rightarrow$ two equal base angles
- Common Pythagorean Triples: 3,4,5—5,12,13—7,24,25

Polygons

- Interior angle sum = $(n - 2) \cdot 180^\circ$
- Each interior angle (regular polygon) = $[(n - 2) \cdot 180^\circ]/n$
- Each exterior angle (regular polygon) = $360^\circ/n$

Quadrilaterals

- Parallelogram area = bh
- Trapezoid area = $\frac{1}{2}(b_1 + b_2)h$
- Diagonal of rectangle = $\sqrt{l^2 + w^2}$

3D Figures

- Rectangular prism diagonal = $\sqrt{l^2 + w^2 + h^2}$
- Cylinder surface area = $2\pi r^2 + 2\pi rh$
- Cone surface area = $\pi r^2 + \pi rs$ (s = slant height)

Coordinate Geometry

- Distance formula = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Midpoint formula = $((x_1 + x_2)/2, (y_1 + y_2)/2)$
- Slope of perpendicular lines = negative reciprocal

Special Notes

- Diagonals of a square = $\text{side}\sqrt{2}$
- Diagonals of a rhombus are perpendicular and bisect each other
- Area of equilateral triangle = $(\sqrt{3}/4)a^2$



EXPONENTIAL EQUATIONS

Exponential Equations:

Exponential equations involve a variable in the exponent, typically written as $y = a(b)^x$, where:

- a: Initial value
- b: Growth/decay factor / percent
- x: Exponent (time or number of intervals)



For example, if Chiara's hair is 20 inches long, and it grows 6% every week, we could use this equation: $l = 10(1.06)^w$, where l is length and w is the number of weeks of growth.

Compound Interest:

A common exponential formula is compound interest:

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

Where:

- A: Final amount
- P: Principal (starting amount)
- r: Annual interest rate (as a decimal)
- n: Number of times interest is compounded per year
- t: Time in years



So if Bob put \$5,000 into an account that earns 3.5% annual interest, compounded quarterly, we could use this equation: $A = 5,000(1.00875)^{4t}$

EXPONENTS & RADICALS

Exponents

- Product Rule: $a^m \times a^n = a^{m+n} \rightarrow x^3 \times x^4 = x^7$
- Quotient Rule: $\frac{a^m}{a^n} = a^{m-n} \rightarrow \frac{x^9}{x^3} = x^6$
- Power Rule: $(a^m)^n = a^{m \times n} \rightarrow (x^3)^4 = x^{12}$
- Distributing Powers: $(ab)^n = a^n \times b^n \rightarrow (2x)^3 = 2^3 \times x^3 = 8x^3$
- Zero Exponent: $a^0 = 1 \rightarrow 4^0 = 1$
- Negative Exponent: $a^{-n} = \frac{1}{a^n} \rightarrow x^{-2} = \frac{1}{x^2}$

Radicals

- Fractional Exponents: $a^{\frac{m}{n}} = \sqrt[n]{a^m} \rightarrow x^{\frac{7}{3}} = \sqrt[3]{x^7}$
- Simplifying Radicals: $\sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2}$



FUNCTIONS & TRANSLATIONS

Function Notation

- $f(x)$ = expression $\rightarrow$ replace x with the input
- Example: $f(x) = 2x + 3$
 - $f(4) = 2(4) + 3 = 11$
- Example: $g(t) = t^2 - 5t \rightarrow g(-2) = (-2)^2 - 5(-2) = 4 + 10 = 14$

Composite Functions

- $(f \circ g)(x) = f(g(x))$
- Example: $f(x) = x + 1, g(x) = 2x$
 - $(f \circ g)(3) = f(g(3)) = f(6) = 6 + 1 = 7$

Transformations of Graphs

- $f(x) + c \rightarrow$ shifts up c units
- $f(x - c) \rightarrow$ shifts right c units
- $f(x + c) \rightarrow$ shifts left c units
- $-f(x) \rightarrow$ reflection over the x -axis
- $f(-x) \rightarrow$ reflection over the y -axis
- $af(x)$:
 - If $|a| > 1 \rightarrow$ vertical stretch
 - If $0 < |a| < 1 \rightarrow$ vertical shrink

Example: $f(x) = x^2 \rightarrow g(x) = (x - 2)^2 + 3$

- Shift right 2, up 3

RATIOS & PROPORTIONS

Ratios

- Written as $a:b$, a/b , or “ a to b ”
- Can be scaled up or down: $3:5 = 6:10 = 9:15$

Proportions

- Equation stating two ratios are equal
 - Example: $2/3 = 4/6 \rightarrow$ true proportion
- Cross multiply to solve:
 - If $a/b = c/d \rightarrow ad = bc$

Scale Factors & Similar Figures

- In similar shapes, sides are proportional
- Scale factor = (new length)/(original length)
 - Example: A triangle with sides 3, 4, 5 is enlarged by scale factor 2 $\rightarrow$ new sides 6, 8, 10

Direct Variation

- $y = kx$ (k = constant of proportionality)
- As one increases, the other increases proportionally
- Example: If $y = 5$ when $x = 2$, then $k = 2.5 \rightarrow$ equation $y = 2.5x$

Inverse Variation

- $xy = k$ (product is constant)
- As one increases, the other decreases
- Example: If $x = 4$ and $y = 6 \rightarrow k = 24$.
 - When $x = 8, y = 24/8 = 3$



STATISTICS

Measures of Central Tendency

- Mean (average): $\text{sum} \div \text{number of values}$
 - Example: $(5 + 7 + 8)/3 = 20/3 \approx 6.67$
- Median: middle value when data is ordered
 - Example: $[3, 5, 8, 9, 12] \rightarrow \text{median} = 8$
- Mode: most frequent value
 - Example: $[2, 4, 4, 5, 7] \rightarrow \text{mode} = 4$

Measures of Spread

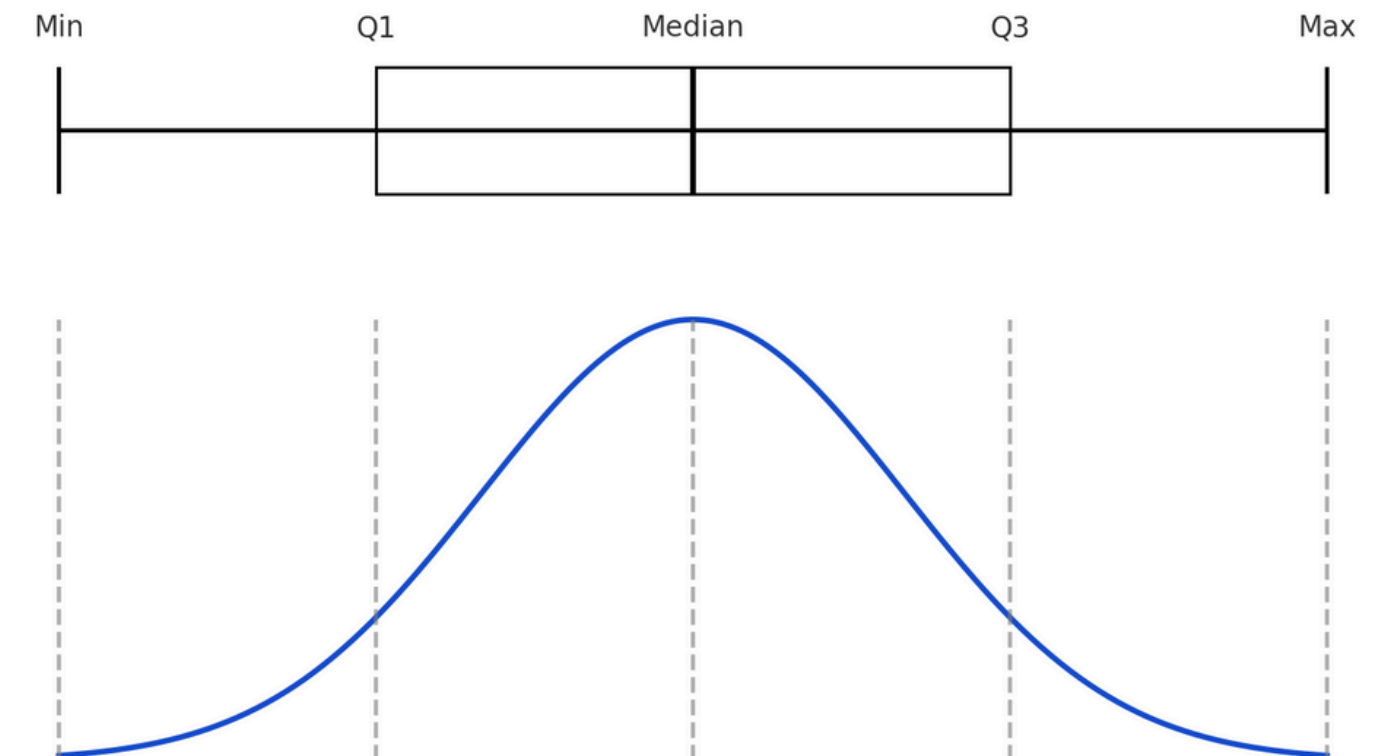
- Range: $\text{max} - \text{min}$
 - Example: $[3, 5, 8, 12] \rightarrow \text{range} = 12 - 3 = 9$
- Interquartile Range (IQR): $Q3 - Q1$ (middle 50% of data)
- Standard Deviation (SD): measure of spread around the mean
 - Large SD $\rightarrow$ values spread out
 - Small SD $\rightarrow$ values clustered near the mean

Sample vs. Population

- Population: entire group (all students, all voters, all products)
- Sample: smaller group meant to represent the population
- To apply sample info to a population:
 - Sample must be random and representative
 - Same type of group $\rightarrow$ don't generalize from adults to teenagers, or from one city to the whole country
- Trap: Absolute language ("this proves," "always," "never") $\rightarrow$ usually wrong
- Correct SAT wording: suggests, indicates, may reflect

BOX & WHISKER

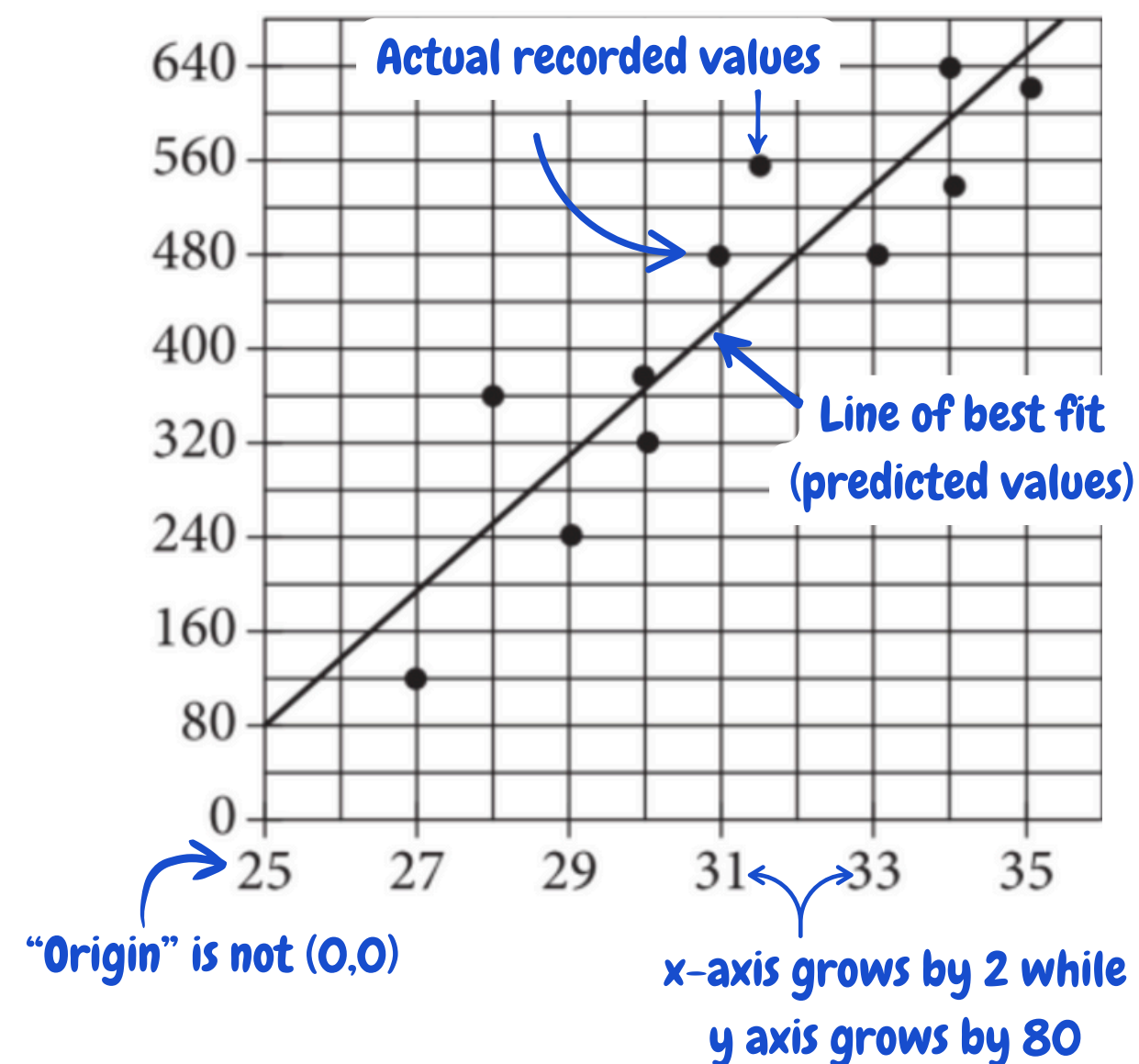
Box & Whisker Plot aligned with Bell Curve



- Divides data into 4 equal parts (quartiles).
- **Median** = middle line; **Range** = $\text{Max} - \text{Min}$.
- Each quartile = 25% of data.
- **Right Skew** (longer right whisker, tail to the right):
 - More extreme high values.
 - $\text{Mean} > \text{Median}$.
- **Left Skew** (longer left whisker, tail to the left):
 - More extreme low values.
 - $\text{Mean} < \text{Median}$.



SCATTERPLOTS / LINE OF BEST FIT



- Line is an approximation—it doesn't pass through all points.
- Watch for axis tricks:
 - "Origin" may not be (0,0), which can affect what we think the y-intercept is
 - X and Y axes may grow at different rates, making the slope look steeper or flatter than it really is.

PROBABILITY

Probability = Desired Outcomes ÷ Total Outcomes.

- Can be written as fractions, decimals, or percents.
- "AND" means both events must happen.
 - Multiply probabilities (if independent).
 - Example: Rolling a 2 and then a 5 on a die: $(1/6) \times (1/6) = 1/36$
- "OR" means either event can happen.
 - Add probabilities (subtract overlap if not mutually exclusive).
 - Example: Rolling a 2 or a 5 on a die: $1/6 + 1/6 = 2/6 = 1/3$

Mutually Exclusive Events:

- Events cannot both happen at the same time.
 - Example: Getting a 2 and getting a 5 on a die roll are mutually exclusive.

Independent Events:

- One event does not affect the other.
- Example: Flipping a coin and rolling a die.

Dependent Events:

- One event changes the probability of the other.
 - Example: Drawing a card without replacement.

Common Mistakes:

- Confusing "AND" with "OR."
- Forgetting to subtract overlap for "OR" problems when events are not mutually exclusive.
 - Example: Drawing a card that is a red card **or** a face card
 - Mistake: $26/52 + 12/52 = 38/52$ — Correction: Subtract overlap (red face cards = $6/52$). Correct total = $32/52$
- Assuming independence when the events are dependent.
 - Example: Drawing 2 aces from a deck without replacement
 - Mistake: $4/52 \times 4/52$ — Correction: After drawing one ace, only 3 remain. Correct: $4/52 \times 3/51$



FACTORING PATTERNS

Difference of Squares

- $a^2 - b^2 = (a - b)(a + b)$
 - Example: $x^2 - 25 = (x - 5)(x + 5)$

Perfect Square Trinomials

- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a - b)^2 = a^2 - 2ab + b^2$
 - Example: $(x + 4)^2 = x^2 + 8x + 16$

Sum & Difference of Cubes

- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
 - Example: $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$

MATH WITH POLYNOMIALS

Adding & Subtracting Polynomials

- Combine like terms (same variable, same exponent).
- Example: $(3x^2 + 2x + 5) + (2x^2 - 7) = 5x^2 + 2x - 2$
- Example: $(4x^3 - 6x + 1) - (2x^3 + 3x - 8) = 2x^3 - 9x + 9$

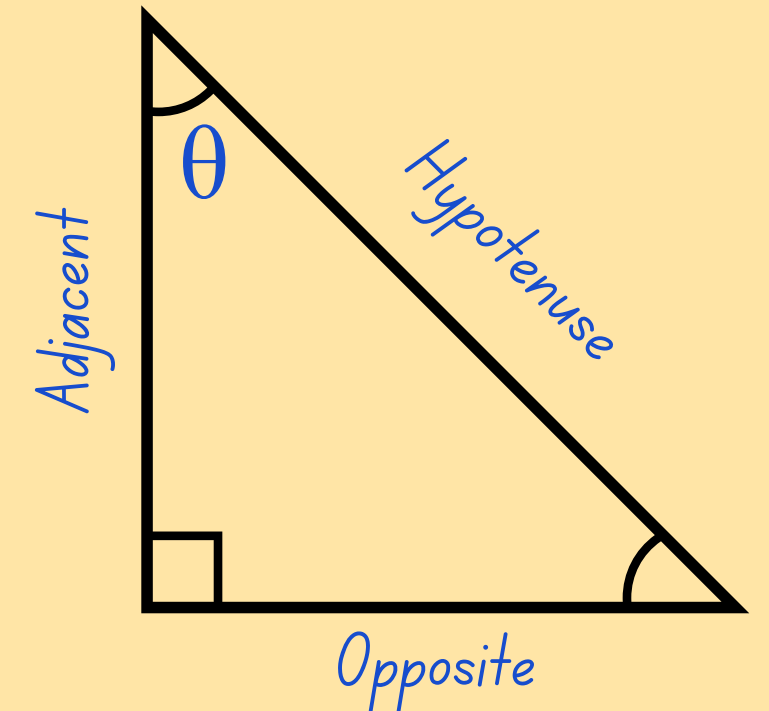
Multiplying Polynomials

- Distributive Property / FOIL (First, Outer, Inner, Last).
- Example: $(x + 3)(x + 5) = x^2 + 8x + 15$
- Example: $(2x - 1)(x^2 + 4x + 3) = 2x^3 + 8x^2 + 6x - x^2 - 4x - 3 = 2x^3 + 7x^2 + 2x - 3$

TRIGONOMETRY

Right Triangle Ratios (SOH-CAH-TOA)

- Sine θ = Opposite / Hypotenuse
- Cosine θ = Adjacent / Hypotenuse
- Tangent θ = Opposite / Adjacent



Radians & the Unit Circle

- $180^\circ = \pi$ radians
- 1 full circle = 2π , but angles keep going: $2\pi = 360^\circ$, $4\pi = 720^\circ$ (two full circles), $6\pi = 1080^\circ$ (three full circles)
- Key conversions: $30^\circ = \pi/6$, $45^\circ = \pi/4$, $60^\circ = \pi/3$, $90^\circ = \pi/2$
- Any angle plus a multiple of 2π lands in the same position.
 - Example: $\pi/4$, $9\pi/4$, $17\pi/4$ are all the same location.

